Lesson 6: Add a timer

Pygame CE • S3 onwards • 50-60 min

Learning intention

Use elapsed time to create a timed challenge.

Success criteria

- I can calculate elapsed time.
- I can display a countdown.
- I can stop scoring when time runs out.

Key ideas

- **pygame.time.get_ticks()** gives milliseconds since pygame started.
- Remaining time = round length - (now - start time).
- `max(0, ...)` stops the countdown going negative.

Activities

1. Decide how long a round lasts and why.
2. Display the remaining time.
3. Prevent scoring when the time reaches zero.
4. Playtest with a partner and adjust the length.

Code

Key changes

```
start_ticks = pygame.time.get_ticks()

# inside the loop:
elapsed = (pygame.time.get_ticks() - start_ticks) / 1000
remaining = max(0, 30 - elapsed)

if remaining > 0 and player_rect.colliderect(target):
    ...

time_text = font.render(f"Time: {remaining:.0f}", True, (255, 255, 255))
screen.blit(time_text, (10, 45))
```

Complete program

```
import pygame
import random

pygame.init()

WIDTH = 800
HEIGHT = 600
SIZE = 40

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Timed challenge")
clock = pygame.time.Clock()

x = 380.0
y = 280.0
speed = 250
target = pygame.Rect(random.randint(0, WIDTH - SIZE),
                     random.randint(0, HEIGHT - SIZE), SIZE, SIZE)
```

```
score = 0
font = pygame.font.Font(None, 36)
start_ticks = pygame.time.get_ticks()
running = True

while running:
    dt = clock.tick(60) / 1000

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    x += (keys[pygame.K_RIGHT] - keys[pygame.K_LEFT]) * speed * dt
    y += (keys[pygame.K_DOWN] - keys[pygame.K_UP]) * speed * dt
    x = max(0, min(x, WIDTH - SIZE))
    y = max(0, min(y, HEIGHT - SIZE))

    elapsed = (pygame.time.get_ticks() - start_ticks) / 1000
    remaining = max(0, 30 - elapsed)

    player_rect = pygame.Rect(int(x), int(y), SIZE, SIZE)
    if remaining > 0 and player_rect.colliderect(target):
        score += 1
        target.topleft = (random.randint(0, WIDTH - SIZE),
                          random.randint(0, HEIGHT - SIZE))

    screen.fill((30, 30, 40))
    pygame.draw.rect(screen, (240, 190, 60), target)
    pygame.draw.rect(screen, (80, 200, 120), player_rect)
    score_text = font.render(f"Score: {score}", True, (255, 255, 255))
    screen.blit(score_text, (10, 10))
    time_text = font.render(f"Time: {remaining:.0f}", True, (255, 255, 255))
    screen.blit(time_text, (10, 45))
    pygame.display.flip()

pygame.quit()
```

Checkpoint

The timer counts down to zero and no more points can be scored.

Support and stretch

- **Support:** Provide the display code already placed; learners add the calculation.
- **Stretch:** Add a time-bonus target, or let the player choose the round length.

Reflect

Is 30 seconds the right difficulty? What evidence do you have?

Save your work
Save this version as `lesson6.py` before you start the next lesson.